

Related Work: Malware Image Classification with CNN and Attention

Malware binaries can be represented as grayscale or color images so that recurring byte-level patterns can be learned by image-classification models. The studies below were selected because they use convolutional neural networks, channel or spatial attention, cross-modal attention, or multi-head self-attention on Maling or closely related malware-image settings. The review focuses on the model design, attention mechanism, evaluation protocol, reported result, and limitation relevant to the assigned Custom CNN + Attention track.

1. Van Dao et al. (2022) [1]. proposed a lightweight feature-synthesis pipeline that combines a small CNN with an attention-enhanced variational autoencoder. The attention component follows CBAM principles by using channel and spatial attention to refine the learned representation. The extracted features were classified with several conventional classifiers; Random Forest produced the best reported accuracy of 99.40% on the unbalanced Maling dataset under 10-fold cross-validation, while the reduced balanced setting achieved 98.40%. A key limitation for the present project is that the gain cannot be attributed to attention alone because the pipeline combines CNN features, an attention-VAE representation, and an external classifier rather than comparing the same end-to-end CNN with and without attention.

2. Kim et al. (2023) [2]. introduced an attention-based cross-modal CNN that fuses two representations derived from non-disassembled malware binaries: malware images and structural-entropy sequences. Cross-modal attention allows each modality to reinforce the other before higher-level CNN feature extraction. Using 10-fold cross-validation, the model achieved 99.09% accuracy and an F1-score of 0.976 on a filtered Maling set of 8,250 samples. This study demonstrates that attention can improve multimodal fusion, but its input requirements are more complex than an image-only project: original binaries are needed to derive structural entropy, and the performance contribution of attention is mixed with the benefit of adding a second modality.

3. Ben Abdel Ouahab et al. (2023) [3]. evaluated a Vision Transformer for Maling classification and compared it with a conventional CNN. The ViT divides each image into patches and uses multi-head self-attention to capture long-range dependencies. The reported ViT accuracy was 98.38%, compared with 97.90% for the authors' CNN baseline. The work provides a direct CNN-versus-self-attention comparison, but the ViT requires greater computational resources and longer training time. It also does not isolate channel attention, spatial attention, and their combination within the same custom CNN backbone, which is the central comparison required in the present project.

4. Dong (2024) [4]. presented a convolution-free malware-image framework consisting of opcode-to-RGB conversion, GAN-based synthetic oversampling, and a simplified Vision Transformer classifier. Multi-head self-attention is used to model global relationships between image patches. The method achieved 96.97% accuracy and an F1-score of 0.9702 on MALIMG. The approach addresses imbalance and global context, but the preprocessing and GAN augmentation differ substantially from the original grayscale-image setting. Because image conversion, GAN oversampling, transfer/adaptation choices, and self-attention are introduced together, the independent effect of attention is difficult to determine.

5. Ayturan (2026) [5]. integrated a Convolutional Block Attention Module after a pretrained ResNet-50 feature extractor. The CBAM sequentially applies channel attention and spatial attention to emphasize discriminative malware textures. The model achieved 99.36% test accuracy on Maling and 97.25% accuracy on MaleVis, providing useful evidence of cross-dataset generalization. However, the model relies on a large pretrained backbone and targeted augmentation, and it does not provide a controlled ablation using the same lightweight custom CNN for no attention, channel-only attention, spatial-only attention, and full CBAM.

6. Uddin et al. (2026) [6]. proposed ParaECA-LSTMNet, which combines four parallel CNN branches, Efficient Channel Attention, additional convolutional refinement, and an LSTM layer. On Maling, using a stratified 70%/15%/15% train-validation-test split, it reported 99.23% accuracy, 99.23% precision, 99.20% recall, and 99.20% F1-score. The design shows that channel recalibration can complement multi-scale and sequential feature modeling. Nevertheless, several components are changed simultaneously, so the contribution of ECA cannot be isolated cleanly; the reported experiment also does not replace the need for five-fold cross-validation and significance testing against a matched CNN baseline.

Summary Table:

Paper	Year	Dataset / Protocol	Method	Attention Type	Main Result
Van Dao et al.	2022	Malimg; 10-fold CV; unbalanced and reduced balanced settings	Lightweight CNN + attention-VAE features + Random Forest	CBAM-style channel + spatial attention	99.40% accuracy on unbalanced Malimg; 98.40% on reduced balanced data
Kim et al.	2023	Filtered Malimg (8,250); 10-fold CV	Cross-modal CNN using malware images + structural entropy	Cross-modal attention	99.09% accuracy; F1 = 0.976
Ben Abdel Ouahab et al.	2023	Malimg; ViT compared with authors' CNN	Vision Transformer	Multi-head self-attention	ViT 98.38% accuracy; CNN 97.90%
Dong	2024	MALIMG; opcode-to-RGB images with GAN oversampling	Simplified ViT + GAN + adaptive/transfer-learning pipeline	Multi-head self-attention	96.97% accuracy; F1 = 0.9702
Ayturan	2026	Malimg test partition + MaleVis cross-dataset evaluation	Pretrained ResNet-50 + CBAM	Channel + spatial attention	99.36% on Malimg; 97.25% on MaleVis